

“It’s Time!” Toward a Human-AI Quantum Experience Design Paradigm: Reinventing the Theoretical Framework of HCI

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ABSTRACT

In an AI-driven digital age, it appears that traditional Human-Computer Interaction (HCI) theories and paradigms need to be reassessed, as they should be realigned and augmented to better serve a broader purpose and context. Current Human-Centered Design (HCD) approaches lack the flexibility, volatility and openness to accommodate the new requirements of a highly demanding intelligent digital reality, like randomness, uncertainty, multi-purpose dynamic interactions, adaptability, data-driven design and decision-making, etc. In this respect, this paper (a) embraces an alternative Quantum User Experience theoretical stance – by creating parallels to the fields of (Quantum) Physics, to facilitate the understanding of HCI phenomena in a blended digital-physical ecosystem; and (b) proposes a Human-AI Quantum Experience Design Paradigm that focuses on the dynamics generated by the respective micro- and macro-experiences. Accordingly, the 5 phases: Explore. Define. Engineer. Transform. Adopt., are outlined and brought into perspective of an extended HCD scope. Main aim is to present a promising theoretical standpoint to the HCI community that might be able to address core theoretical challenges and discrepancies, and complement HCD practices towards inclusive and sustainable human experiences.

CCS CONCEPTS

• **Human-centered computing** → **HCI theory, concepts and models**; • **Computing methodologies** → **Artificial intelligence**;

KEYWORDS

HCI, Quantum User Experience, Human-centered Artificial Intelligence, Paradigm, Quantum Computing

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1 INTRODUCTION

Nowadays, it is quite evident the importance of Human-Computer Interaction (HCI) and User Experience (UX) in the design and development of systems and applications. Especially, if they need to operate in a demanding digital ecosystem that is characterized by (a) extensive utilization of upgraded cloud-based infrastructures [31]; (b) zillions of data – exponential growth of multivariate data sets [2]; (c) profound advancement and integration of Artificial Intelligence (AI) models (from pure analytic and predictive to generative) to the core of solutions [22]. From the humans standpoint, the current rich technological reality creates diverse expectations and requirements, as users create new subjective and social experiences during their interactions in highly interconnected and in situations blended realities. In fact, the interactions per se, are now in the center of attention since their nature changes, becoming even more difficult to specify [18, 20]. For example, they are fluid, multi-modal and ubiquitous; compiled in real-time, based on context and relevance; intent-driven entailing various devices and contexts of use for a single goal; and they may not necessarily follow a deterministic success flow.

In current times, it seems that traditional HCI models and practices, like Human-centered Design (HCD)¹ [21], User-Centered Design (UCD) [1], Double Diamond [8], Design Thinking [5] etc., that guide the creation of new contents, interactions and designs of applications or services, lack the flexibility to comply with the continuously changing technological dynamics, complexity and requirements [10, 25]. Main weaknesses focus on their by definition sequential nature – with demanding structural transitions between the various phases; rigid and subjective pre-defined interaction conditions (for situation-specific use cases), without the potential of adaptive scenarios; long feedback cycles with a significant time gap from feedback-generation to feedback-integration; not data-driven oriented, e.g., there is a high risk that user input received from algorithms or computational channels not to be proactively considered in the HCD process (e.g., during the Exploration or Discover phases) or to be loosely integrated after development. In particular, with the intense presence of AI in applications and services, HCI researchers and professionals would need to employ human-centered AI design approaches [3, 34] rather than the traditional ones, for acknowledging according to Schneiderman “humans in the group; computers in the loop” [30]. In such a case, the respective AI capabilities, uncertainty and constraints would be considered early throughout the HCD process (e.g., handling complex outputs in

¹ISO 9241-221:2023 standard: <https://www.iso.org/standard/81384.html>

relation to prototypes design or evaluation with real users) [35], designing solutions by bringing out the best of human ability, rather than simply attempting to automate them away (i.e., human-AI complementarity [17]).

As technology advances, enhancing our physical environment and amplifying users' real-life actions and natural activities, a crucial question is which theoretical principles would be able to augment HCI theory and practices? Taking a closer look to the problem, we may recognize that it presents similarities to research questions that relate to phenomena that define nature (or life)². Hence, in this paper we discuss an alternative theoretical perspective to HCI, by embracing principles of disciplines that study the physical world, like Physics and more specifically the research directions of Cosmology (e.g., Theory of Relativity [26]), Quantum Physics [11] (or Mechanics), and complementary String Theory [37]. More specifically, we propose a: (a) Quantum User Experience (QUX) theoretical stance, using an analogy to our physical world and the respective theories³; and (b) Human-AI Quantum Experience Design (HAI-QXD) paradigm, that is composed of 5 interrelated dimensions: Explore. Define. Engineer. Transform. Adopt. We believe that the current proposal may offer an approach to HCI community, as a research direction, to build upon for satisfying the new expectations and requirements, and explain HCI phenomena in an unpredicted blended digital-physical reality that changes faster than ever.

2 OUR DIGITAL COSMOS THROUGH AN ALTERNATIVE QUANTUM UX THEORETICAL LENS

In light of a more holistic thinking, we argue that our digital cosmos (or world) as a global network (of networks, i.e., the Internet), should be recognized on two scales, i.e., macroscopic and microscopic, depending on the reference frame of an observer existing and functioning in it. A *macroscopic* view refers mostly to observations of phenomena that occur in this network (also in relation to others), and we can detect, analyze, and influence at a large-scale (i.e., “from a distance”). On the contrary, a *microscopic* view (i.e., small-scale) defines what can be perceived and felt but not directly “see them with a naked eye”. The two views (or realization levels) co-exist and affect one another (parallel: Schrödinger’s Cat paradox⁴ [33]), with micro-phenomena to be extended into macro-phenomena. Fundamentally, we assume that our digital cosmos is composed of three entities (for the purpose of this paper we consider a certain level of abstraction): The *Agents* (i.e., users, although an inclusive scenario would relate to any entity able to interact with Objects or with other Agents, like intelligent bots, engines, platforms, animals, etc.), *Interactions* (i.e., any exchange or communication between Agents, devices, or entities that occurs through the digital medium), and *Objects* (e.g., any type of content, like text, audio, visual, video,

etc.)⁵ As Agents engage in the digital cosmos with Objects or other Agents to meet their goals, they generate fluent (user) experiences assembled by the respective interactions, that may be characterized by a duality, conceived as: Macro-experiences or micro-experiences. *Macro-experiences* refer to collective behavioral patterns in relation to defined reference points to the data sphere (global perspective). They may involve multiple entities, have a substantial impact on an individual or group, and be formulated in more extended periods of time. On the contrary, *micro-experiences* may be acknowledged as individual perceptual responses to stimuli (local perspective), that are brief (often seconds or moments) in relation to notable events that might be important for an individual or seem insignificant on their own but collectively shape the overall perceptions and feeling of the individual. They both contribute to the overall, cumulative, experience (or QUX) and information that continuously flow across the structures and billions of nodes of the global network, composing a huge data field that comes into different shapes and forms, i.e., the “data fabric of the digital cosmos” (parallel: we coin the term inspired by the fabric of the cosmos [15]). It reflects the interconnected and dynamic nature of the digital reality (both on a cosmological and quantum scales), that is defined within the boundaries of time (When), space (Where), and texture (What – see conceptual Figure 1A – macro-experience angle), and gives rise to the observable phenomena consisting of the behavior of Agents, Objects and Interactions (as the yielded Quantum (User) Experience Energy (QXE) – parallel: the universe is not a void space [32]⁶).

Continuing this analogy, key theoretical concepts of the digital cosmos and fundamental QUX components (see Table 1) may relate to, in a nutshell: (a) Digital Cosmic Web: A vast network-like structure where nodes and nodes clusters are interconnected by digital cosmic threads⁷ (e.g., short- or long-term connections between the Agents and Objects); (b) Space-Time Continuum: All interactions (QXE) between Agents and Agents with Object(s) take place within this continuum curving the space-time data fabric (parallel: Einstein’s theory of general relativity [4]⁸); (c) Fundamental Forces: The data fabric of digital cosmos is influenced by fundamental forces (parallel: e.g., gravity) such as inter-connectivity (e.g., engagement), experiences (both macro- and micro-), data (e.g., the volume of data) an Object may have, determining, e.g., how “viral” a node might be; (d) Quantum Digital Fields: The data fabric of the digital cosmos is also associated with pervasive digital fields that exist throughout space and give rise to fundamental elements, e.g., human traits, described as qubits⁹ (parallel: particles, their properties, and their interactions [24]¹⁰).

⁵Content may be authentic/ human-generated or synthetic/ AI-generated, and may be available as standalone or part of other broader entities like systems, applications, Web-sites, etc. For simplicity we will consider content providers as part of this entity.

⁶Universe is a dynamic and interconnected entity with a structure that governs the behavior of matter and energy.

⁷Parallel: the matter in our contemporary universe is organized into an extensive network of filamentary structures, i.e., cosmic threads [6].

⁸Gravity is not just a force between masses but a curvature of this space-time fabric caused by the presence of mass and energy.

⁹Referring to a quantum system – as opposed to traditional bits 1 and 0.

¹⁰Using quantum mechanics as a framework that governs their behavior on a very small scale. In reality a human at a quantum/ subatomic scale could be regarded as a particle (i.e., made of cells - molecules - atoms - particles), but for the scope of this paper we keep it as a parallel, so as to make more clear the distinction of the digital cosmos and respective elements.

²The fundamental principles of the universe, its composition, motion, and behavior through space-time; or on very small scales the behavior of the matter and energy, such as atoms and subatomic particles.

³For simplicity, we will use the word “parallel” for an explicit reference that conveys a metaphorical theoretical association, whenever necessary, avoiding lengthy descriptions due to space limitation. In addition, an exhaustive association of the subsequent theoretical components or deep-dive into specifics is out of this paper’s scope – and a rather ambitious endeavor at this stage.

⁴Illustrating the strange implications of superposition; elements exist and do not exist until they are observed or measured (observer-dependent reality).

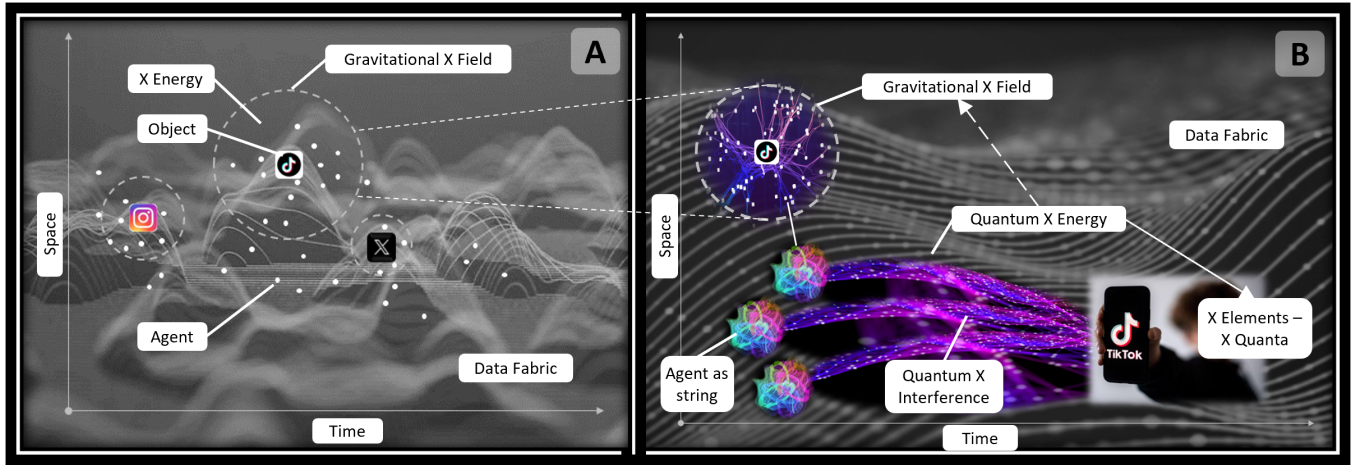


Figure 1: Macro- & Micro-Experiences (X) angle of the Data Fabric of Digital Cosmos: (A) A more holistic, large-scale perspective of an observer, (B) Zoom-in to the Gravitational Experience Field of an Object and small-scale quantum UX angle.

Table 1: Fundamental components of Quantum UX.

QUX Component	Description
Agent	User (atom) as a string of x vibrational patterns (e.g., individual differences – human traits)
Object	Content composed of various types of (Experience) Elements, which are specified as a probabilistic distribution and associated with, e.g., UX processes, XQ, ES
Gravitational Exp. Field (GXF)	A region in the digital space around a “massive” Object, such as a highly popular app, where another Object experiences a force due to gravity (i.e., highly concentrated energy levels (XQ))
Experience Elements (XL)	The building blocks of an Object. They are dynamic (able to be shaped and reshaped), reusable, multidimensional elements (e.g., buttons, or submission controls) that exist in multiple forms (e.g., shaping up the structure, look and feel of an Object, e.g., Online service)
Quantum Exp. Energy (QXE)	Interaction states producing, e.g., feeds, posts, (re-) actions, feedback. They set the XL of an Object into motion (e.g., keep an app alive), and they are characterized by intensity and variance
Experience Quanta (XQ)	Refer to discrete packets or units of energy (i.e., QXE) associated with the behavior of XL – nature of certain properties (e.g., the property color, shape or orientation of an element)
Excited State (ES)	Refer to the (temporary) higher energy states of Elements (e.g., XQ of an Object) than their baseline (e.g., more likes)

For example, Figure 1A depicts an instance of the data fabric of the digital cosmos as a continuously changing fundamental structure of interconnected data among numerous Objects (Experience Elements (XL) – e.g., social media applications like TikTok, Instagram, and X¹¹). We can observe that the current form of the digital fabric is “wavy”, consisting of smaller or bigger entities’ (or Objects’) “energy hills” (Experience Quanta (XQ) and Excited States (ES)), as the data (e.g., information flow) for each Object is generated at a specific space-time interval via Agents’ interactions (QXE). Following the Agents’ engagement and energy levels, each Object has a specific “mass” (e.g., number of users, volume of information) that determines its Gravitational Experience Field (GXF – see Figure 1B – zooming-in an observer’s small-scale perspective to the digital cosmos). This GXF is able to influence the behavior of Agents

for a specific Object (e.g., due to popularity or social desirability bias [16]) that inevitably defines its dependency with the other Objects (e.g., competitive advantage, or shares in the market), how they are situated on the data fabric, wrapping space-time. These relationships among the Objects is in constant change, defining an ecosystem that is found in an equilibrium [14], but without being able to predict its values [9], e.g., what would be the “mass” and GXF texture (e.g., Objects’ XL, XQ - energy levels) in the future. Nevertheless, how is the GXF being created and maintained in time? How do users generate QXE fused into objects, setting them into motion? How do the XL receive QXE?

3 A HUMAN-CENTERED QUANTUM REALM

To address these questions, we would need to investigate how Agents, Interactions and Objects function also on a Quantum level (small-scale).

¹¹For the scope of this example we make the assumption that, TikTok (<https://www.tiktok.com/en/>) has approx. over 2 billion downloads worldwide, Instagram (<https://www.instagram.com/>) has approx. over 1 billion monthly active users, and X (<https://twitter.com/>) has approx. over 330 million monthly active users worldwide.

3.1 Agents as Subatomic Oscillating Strings

An Agent (e.g., person-user) may also be regarded as an entity that can be determined on two levels: (a) *Macroscopic*, that entails all the aspects and characteristics that we can observe or receive as an action or reaction of a user, and drive a behavior on a given stimuli or motivation; and (b) *microscopic*, that refers to intrinsic “elementary” human traits, static or dynamic (e.g., cognitive, social, psychological), that are defined within the boundaries of a user’s soma (i.e., body, as an indivisible “atom”) and their value differ between individuals¹². The user (atom) can be considered as elementary only in relation to all individualistic properties (microscopic), but not to the observable (macroscopic) ones such as bodily, behavioral, skills, actions that participate in creating reality, etc. [13] In a microscopic reality (“subatomic”), a user reflects a “string” (parallel: the viewpoint of String Theory [37]¹³). A string may fundamentally conceptualize a person as an one-dimensional vibrating and extended entity, that defines its nature (as atom) with properties and attributes as those defined by the Theories Individual Differences [7], and triggers a behavior. On macroscopic scales larger than the string scale, a string will look just like an ordinary atom (i.e., person), with all the “visible” characteristics determined by the vibrations of the string (and the associated values of its properties in a microscopic scale). QUX builds on the premise that everything that we observe in the digital cosmos (at a macroscopic scale) ultimately stem (directly or indirectly) from the various vibrational states of these fundamental strings (person-users) at a microscopic scale.

3.2 Interactions as Quantum (User) Experience Energy

Interactions reflect the behaviors between the Agents (i.e., acting strings) or Agents and Objects that might be unique or repeated in time producing QXE. The respective energy consists of a variety of interactions driven by, e.g., users’ perceptions and experiences over time (see Figure 1B – energy colour), which entail a certain degree of uncertainty and reflect a probabilistic distribution in the quantum field (parallel: Quantum Field Theory [27]). There are not definite interactions between the involved entities but rather numerous distinct states of those interactions that exist at the same time (parallel: expressed as wave functions), and carry properties like intensity, i.e., the strength of interaction (e.g., positive input, feedback), and variance, i.e., frequency that they occur, and may determine how consistent the engagement is during an interaction process. Hence, a system (described as quantum system of interactions) can exist in multiple states or configurations at the same time (parallel: superposition principle¹⁴). The actual state of the system is determined only upon measurement, whereby it “collapses” into one of the possible states (parallel: the wave function of this state collapses to a classical state). For example, several interactions/

processes modeled in certain ways based on statistical probabilities, e.g., “producing” those that carry more energy than the baseline interaction, so as to determine the real value of an engagement. From a QUX perspective, this mind-shift grants the flexibility to welcome all possible interactions during an engagement, by numerous users (i.e., string oscillations), and by applying, e.g., probabilistic inference, to extract the “fittest”, e.g., interactions found in ES. This outcome as QXE (e.g., XQ), would determine at the next stage the energy levels of, e.g., the XL of an Object.

3.3 Objects as Reusable, Multidimensional Experience Elements

An Object is primarily composed of the XL (e.g., visual controls, buttons), which are the building blocks of, e.g., system, application, service. As these come in various forms and dimensions - entailing an inherent dynamicity, randomness and uncertainty about their state over time - they are situated/ specified following a probabilistic distribution (or pattern) in the Object. This implies that they are not fixed or deterministic (having a precise definite state, properties or values), but instead there is a range of possible values associated with each XL and their properties, governed by probabilities, and reflecting the discrete energy units, i.e., XQ, which determine their behavior and state in the Object. The XQ are charged with energy (i.e., discrete quantities) by the data they receive from multiple interaction/ information flows and are able to interact and influence other elements, steering an Object’s motion. For example, the “color: green” and “size: big” of a button are values that have been assigned as a likelihood of a probabilistic pattern, fused with data from XQ that were generated during multiple interactions of k users with the object over a period of time. Rather than making firm predictions, QUX is concerned with the probabilities for potential measurement outcomes, recognizing a more realistic representation of uncertainty in many real-world scenarios. In addition, XL need to be reusable, promoting consistency and user interface (UI) responsiveness. The UI components, (responsive) design patterns, typography and color styles, layout grids, etc., should be versatile, efficient, and easily adaptable, across different contexts or scenarios¹⁵. In such a case, where the changes are continuous, the possibility of multiple elements’ views, interaction flows and Objects’ states could be shaped or reshaped in real-time given the QXE levels and the necessary computation, a corresponding evaluation method should be in place able to assess the performance of different variations of an Object or a set of specific XL that compose a situation-specific use case (e.g., Multivariate Test, as opposed to A/B testing¹⁶).

Figure 1B shows an end-to-end example scenario around the Object TikTok – how information/ energy circulates in the digital cosmos. Agents (e.g., persons-users as strings of x vibrational patterns) interact with the Object TikTok at the same time interval. Users’ interactions with the Object hide an inherent uncertainty, producing for the same use case a set of quantum interactions (multiple states) that exist at the same time. As the user’s subatomic properties take different values each time, the same interactions may be

¹²The human traits dimensions and their association define an excessive complex system; rather impossible to capture, combine or understand as a whole across time (this complexity and multidimensionality may remind an NP-hard computational problem [19]).

¹³We can imagine a string as a small loop or segment of ordinary string (rather than particles), that can oscillate in different ways, taking various different forms and states over time that we cannot model to reason.

¹⁴When two or more waves coincide in space, the resultant disturbance is equivalent to the algebraic sum of the individual disturbances.

¹⁵Methodologies like Atomic Design [12] or Object-Oriented Design [28], may apply in this case as long as they enable elements to be “disturbed” (not to be static) and to exist in multiple ways.

¹⁶Multivariate vs. A/B Testing – Incremental vs. Radical Changes: <https://www.nngroup.com/articles/multivariate-testing/>

expressed in different ways (as a probabilistic distribution) with different intensity (colorful lines, e.g., in the range calm-stressed, for the trait emotionality; or low-high cognitive capacity, for cognitive load; or low-high, with respect to motivation – parallel: Quantum Interference¹⁷) reflecting each time a different truth for the user and its behavior. Measuring the Users-Object engagement (through the quantum interactions that simultaneously exist and not), in the quantum world we have the opportunity to capture all the states conveying the sophistication of the users' nature (e.g., thoughts, feelings, experiences), by, e.g., modeling probabilistic distributions, while interactions are in motion to discover optimum conditions and information like feedback, re-posts, likes, success interaction paths, preferences, etc., that present some interest based on their levels of QXE (e.g., ES – a pick positive reference on specific elements of the Object, like stories scrolling). Next, this energy (charged XQ), is fused to the Object for setting the XL into motion and at a larger extent the Object itself. Finally, the levels of the QXE determine the strength of the GXF of the Object (e.g., it seems currently the Object TikTok features the most frequent and intense interactions in relation to other Objects, with a very strong GXF that makes it relatively popular).

4 A PROPOSED HUMAN-AI QUANTUM EXPERIENCE DESIGN PARADIGM

The proposed HAI-QXD paradigm preserves the QUX interdisciplinary theoretical considerations at its core. It aims to encourage a shift toward a HCD approach in an extended (Quantum) AI-driven purpose and context [23, 36], able to embrace the new HCI expectations and requirements of our digital cosmos. We define HAI-QXD as, *a paradigm that unifies the micro- and macro-experiences of observed HCI phenomena under a coherent framework sufficient to explain and predict most of the behaviors and interactions generated by the various agents of a system*. Figure 2 depicts the conceptual dimensions of the HAI-QXD paradigm, composed of 5 interrelated phases: Explore. Define. Engineer. Transform. Adopt. Its inherent structure has been purposefully crafted in a circular (or better "cloud") form reflecting the multidimensional "quantum", rather sequential, relationship between the main phases, subsequent elements and actions (parallel: an electron in the hydrogen atom¹⁸). This implies that information is fluid under a constant change in shape and weight (there is no strict sequence or fixed state determining rigidly how input/ output flows in time), as phases exist in different locations at the same time around the nucleus (i.e., an Object's QUX insights), until they are physically observed (with the specifics, requirements, and conditions of a product), "collapsing" into a fixed position. Next we outline each phase, with the principles of ongoing evaluation, iterations, adaptation, and improvement inherently engraved at their core.

Explore. Assess the foundations and vision of an AI-driven system (e.g., as product or service). Gather information for understanding new business opportunities, user needs and context, AI potential and added value, and determine the UX scope and

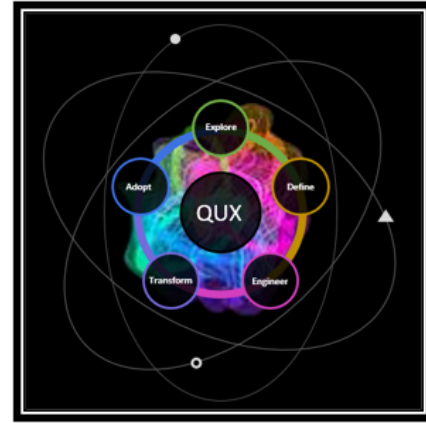


Figure 2: The proposed Human-AI Quantum Experience Design Paradigm (cloud dimensions – solid-circle: UX, empty-circle: AI, triangle: Data).

strategy. The use of existing (meta-) data (i.e., QUX of related Objects) may facilitate the exploration of large sets of qualitative data for identifying patterns, sentiments, and key themes in user responses, providing valuable insights for shaping up a solid basis and expectations; **Define.** Formulate the system's objectives and AI technical specifics, elaborating on the target user group and their related daily activities, synthesizing research findings and clearly articulating the problem of what the system is expected to solve. AI and machine learning techniques can leverage QXE in data-lakes, so as to extract users' behavior based on historical data. Identify relationships, e.g., potential pain points, and provide hidden insights into users' behavior, habits and preferences in relation to specific subatomic elements, like cognitive or emotional traits when interacting with the XL; **Engineer.** Describe the design challenge (solution space), and create the expected (success) user flows enabled by AI capabilities, landscape & possible interaction scenarios. Initialize the system's design layout and elements, define the respective (algorithmic) rules and AI models, and ideate on data-driven concepts, determining specific criteria or parameters that the design solution must meet. Statistical methods (e.g., Bayesian inference) help understand how target users are likely to behave in certain success scenarios (and design concepts), model importance of features and design elements (based on the XL' XQ & ES) and provide predictions on users' future behavior in relation to proposed design concepts (and XL); **Transform.** Building on the power of data, produce design definitions and modeling of multiple (explainable) conditions as adaptable optimum alternatives of the success user flows. A number of different combinations of design interactions based on objectives, constraints, variables and (visual) design guidelines generate new experiences, as candidates for the "survival of the fittest" once executed and receive feedback (QXE) from multiple users over time. AI may facilitate the data-driven design process (e.g., from current UX research, design guidelines, XQ and XL) by applying computational (algorithmic) design techniques like procedural, generative, or combinatorial optimization methods for generating a rich variety of new authentic designs; **Adopt.** Code, integrate and make the system available to the public based

¹⁷When subatomic particles interact with and influence themselves and other particles while in a probabilistic superposition state.

¹⁸The hydrogen atom: https://labs.phys.utk.edu/mbreinig/phys222core/modules/m11/hydrogen_atom.html; colourful string inspired by [29]

on the alternative prototype design flows. Engage into a continuous monitoring of the process, ongoing evaluation and improvement of the system and AI benefits and implications. Emphasis is given to the (re-)modeling and refinement of, e.g., the personalized design conditions, and UX based on user feedback (i.e., QXE on usability, effectiveness, and usefulness) after real-world usage, facilitating the system's (e.g., XL) optimization in time. By taking advantage of the QXE and XQ of XL and with the support of AI and data analytics techniques, the system might be adopted in various ways, e.g., analysis of user behavior and real time interactions may lead to detecting patterns for successful product adoption, insights into user engagement and potential roadblocks.

5 CONCLUSION AND RESEARCH POTENTIAL

A fundamental claim of the QUX and the proposed HAI-QXD paradigm is that everything is measurable, can be computed, combined and change state, supporting the openness and reusability of the produced assets and insights. Hence, in order to maximize its impact, there is the basic assumption that it should be associated with a computational UX infrastructure, perceived as: *The data-driven definition of users' nature and experience by computational thinking and methods that empower adaptability, inclusion, measurability and optimization of designs, interactions and operations.* Although the proposed approach on a theoretical level introduces intriguing opportunities and exciting possibilities for investigation, the practical applicability in real life scenarios would have to address specific challenges and research questions, as future work, that follow its inherent nature and rationale. Those may relate to: How should we explore the concept of superposition in the states of the XL – allowing users existing in multiple states simultaneously? How to embrace the uncertainty principle in UX for creating adaptive interfaces and interconnected experiences? How to acknowledge the inherent uncertainty and probabilistic nature of human choice for modeling a user decision-making processes? How to create global UX that transcend geographical boundaries, involving designing Objects that provide a sense of interconnectedness and shared experiences? How to embrace quantum principles, such as complementarity, allowing users to interact with XL in ways that complement each other, creating richer and more holistic experiences?

In this respect, the main contribution of this work is twofold: On one hand to act as a theoretical framework able to provide an informed basis to explain HCI phenomena given the dynamicity, complexity and uncertainty of current interactions; and on the other hand to serve as a practical guide for the HCI community to spark discussions, an alternative thinking, models and tools, to tackle the challenging design problems of a multidimensional technological reality toward a fair, inclusive and sustainable future for all.

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